

Homework 2

60 points

1. Translate the following into mathematical statements using “implies” notations: **2 + 2 = 4 points**
 - Having a password is necessary for logging in
 - Knowing a formula is sufficient for solving a problem

2. Prove the following using Truth Tables: **6 points**

- $(P \wedge Q) \Rightarrow R \Leftrightarrow P \Rightarrow (Q \Rightarrow R)$

3. Prove the claim in question 2 using De Morgan's Law **6 points**

4. Prove the following using PMI: **10 * 3 = 30 points**

- $2^n > n^2$ for all $n \geq 5$

- $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$ for all $n \in \mathbb{N}$

- Show that $(8^n - 1)$ is divisible by 7 for all $n \geq 1$

5. Prove the following using semantics: **7*2 = 14 points**

- $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$